



OPEN Restoration of aquatic vegetation can mitigate the risk of eutrophication in large shallow lakes

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The role of aquatic vegetation restoration in mitigating lake eutrophication is well-recognized, yet the underlying nutrient dynamics-vegetation response relationship remains poorly quantified, hindering a mechanistic understanding of grass-algae regime shifts. To address this, the Wetland Eco-dynamic Model for Submerged Plants (WET) was developed to mechanistically simulate ecological processes within the water-sediment-plant continuum by coupling hydrodynamics, nutrient cycling, and plant growth modules. Applied to Lake Taihu (2005–2019), the model effectively simulated submerged plant responses to hydrological and physicochemical parameters, with simulated vegetation cover validation achieving an R^2 of 0.68. A multi-scale analysis further identified a significant positive correlation between N:P ratios and aquatic vegetation cover. The model was subsequently employed to simulate vegetation dynamics under 121 distinct nutrient regulation scenarios, revealing that aquatic vegetation cover exerts a stronger influence on total phosphorus than on total nitrogen, a relationship characterized by steady-state nonlinearity. The findings demonstrate that under stable meteorological, hydrological, and pollution load conditions, managing aquatic vegetation is an effective eutrophication mitigation strategy for large, shallow lakes. Specifically for the phosphorus-limited Lake Taihu, simulations indicate that increasing vegetation coverage to 27.7% would shift its trophic status from eutrophic to oligotrophic-mesotrophic. By systematically quantifying these nonlinear responses, this study provides operational thresholds and a scientific basis for lake ecological management, overcoming a key limitation of traditional qualitative descriptions.

Keywords Algal bloom, Submerged plants, State transformation, Threshold mutation, GOTM-FABM-PCLake, Taihu Lake

Eutrophication prevention and control remain a major challenge for global aquatic environmental management¹. This challenge is intensified by increased nutrient loading from anthropogenic activities and climate-induced alterations in hydrodynamics, which have exacerbated the degradation of lake ecosystems^{2,3}. Moreover, controlling pollutant loads alone has proven insufficient for restoring ecological function in many degraded lakes⁴. Consequently, ecosystems often remain impaired by persistent issues such as wetland degradation, eutrophication, and salinization^{5–8}. These impacts pose significant threats to aquatic vegetation, water quality, and public health⁸, thereby compounding the challenges of eutrophication management. This is particularly evident in China, where lacustrine ecosystems have undergone severe degradation due to the combined effects of anthropogenic pressures and natural drivers⁹. Over a recent decade, the proportion of assessed lakes classified as moderately eutrophic increased from 31.3 to 42.7%¹⁰. A rapid shift from mesotrophic to mildly eutrophic status has also been observed in key water bodies, including major tributaries like the Hanjiang River and large reservoirs such as the Three Gorges Reservoir, particularly within their bays and estuaries¹⁰. Nutrient-driven

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eutrophication in Lake Taihu emerged in the 1980s, marked by the expansion of cyanobacterial blooms from Meiliang Bay to the broader northwestern region¹¹. Despite long-standing recognition of the issue, harmful cyanobacterial blooms persist, and anaerobic 'black water' events recur annually each summer^{12,13}. Given that N:P ratios are integrative indicators of nutrient limitation and eutrophic status^{8,14}, a refined understanding of their interaction with biotic components is critical. This study aims to provide this understanding, thereby offering a scientific foundation for the effective management of eutrophication in large, shallow lakes.

Several studies have shown that shallow lakes can shift from a turbid, phytoplankton-dominated regime to a clear-water, macrophyte-dominated regime, indicating a transition from eutrophic to oligotrophic conditions¹⁵. Submerged macrophyte restoration has been widely recognized as an effective strategy for restoring eutrophic aquatic systems¹⁶. The distribution and coverage of aquatic vegetation in Taihu Lake exhibited two distinct phases from 1980 to 2017: a gradual expansion followed by a rapid decline. The distribution area reached its peak between 1980 and 2014, followed by a sharp decrease from 2015 to 2017, declining to less than 50 km². Aquatic vegetation is mainly concentrated in East Lake, the East Area, and Gong Bay, with submerged species constituting the dominant vegetation type¹⁷. Aquatic vegetation plays a critical role in mitigating lake eutrophication through multiple mechanisms, such as suppressing algal growth (Bakker et al., 2010), enhancing dissolved oxygen levels¹⁵, stabilizing aquatic ecosystems¹⁶, and enhancing physical habitat conditions. Beyond these direct interactions, the density and distribution of aquatic vegetation, along with their impact on altering hydrodynamics, also play a crucial role in modifying dissolved oxygen and carbon levels in water, thereby influencing biogeochemical dynamics^{2,3}. A substantial body of research has demonstrated that aquatic plants are key contributors to eutrophication mitigation in lakes^{18,19}. Restoring a balanced macrophyte–algae aquatic ecosystem has become a central focus of ecological restoration efforts in Taihu Lake²⁰. The rehabilitation of eutrophic water bodies using aquatic ecological methods offers a cost-effective and efficient approach to wastewater treatment and is increasingly adopted worldwide to enhance water quality. Lakes dominated by aquatic plants generally exhibit clear water and high transparency under low-nutrient conditions²¹. Notably, clear-water conditions can be maintained in certain lakes through internal regulatory mechanisms, even under elevated nutrient inputs^{22,23}. Conversely, with increasing nutrient inputs, phytoplankton-dominated algal blooms and turbid conditions commonly emerge²⁴. Large-scale aquatic vegetation, including submerged, emergent, and floating-leaf plants, plays a vital role in supporting the structural and functional integrity of lake ecosystems²⁵. Large aquatic vegetation in lakes, particularly submerged species, is widely recognized as an effective means of managing eutrophication in ecological restoration efforts. As primary producers in aquatic ecosystems, submerged plants enhance water quality and reduce eutrophication in aquatic ecosystems²⁶. The transition between clear and turbid water states is primarily driven by competition between submerged plants and phytoplankton. Therefore, the adoption of water ecological restoration methods can better prioritize natural recovery and long-term benefits, overcoming the limitations of short-term solutions and the potential negative impacts of traditional governance approaches. It represents a more sustainable and environmentally friendly governance approach. This approach enhances the self-purification capacity of water bodies, reduces secondary pollution, lowers treatment costs, and demonstrates strong adaptability, aligning with the concept of ecological civilization construction. This paper models the ecological mechanisms of submerged plants in Taihu Lake, quantifying the effects of management measures and providing a scientific basis for formulating specific goals to restore the ecological functions of the lake and restore its clear water state.

Additionally, simulating aquatic vegetation requires comprehensive environmental and biological data, which necessitate complex statistical analysis and model development. Experimental methods often encounter challenges in realistically replicating the evolution of lake ecosystems, while the lack of effective mathematical models hinders the accurate prediction and simulation of the grass–algae transition process. Experimental methods often concentrate on specific species, such as subfossil mosquitoes, zoobenthos, and zooplankton, leading to a highly complex and labor-intensive workload^{27,28}. The impact of aquatic plants on eutrophication can be assessed through experiments involving these plants²⁹. These experiments can be conducted indoors or in small lake areas, which facilitates ease of implementation and reproducibility. However, these experiments are time-intensive and spatially constrained, thereby limiting their representativeness of the entire site and watershed³⁰. Statistical analysis methods are commonly used to identify trends in large datasets over extended periods. These methods are primarily utilized to determine thresholds. However, statistical approaches that do not incorporate mechanistic models to represent the underlying mechanisms of individual components may be susceptible to error propagation during data processing³¹. In contrast to statistical methods, eco-dynamic models can predict changes in lake ecosystems and aquatic vegetation using mechanistic modeling. Several mathematical models are employed to elucidate the mechanisms underlying changes in aquatic ecosystems³². The Vollenweider model was the first to establish the relationship between phosphorus nutrient loading and the eutrophic status of lakes³³. The PCLake (Princeton Lake Model), an integrated model of complex aquatic ecosystems, was used to simulate state changes with delayed effects in bifurcation analyses of a dataset comprising 43 shallow temperate lakes^{34,35}. It has been widely used in studies of regime change, climate change, and submerged plant restoration^{36,37}. The WET model (Water Ecosystems Tool) integrates the hydrodynamic GOTM (General Ocean Turbulence Model) with the PCLake model. Compared to the standalone PCLake model, this coupled model more accurately simulates the gradual response of ecosystems to changes in external forcing. It is particularly useful for studying the responses and changes in aquatic vegetation in shallow lakes, especially in scenario simulations^{38,39}.

However, the quantitative mechanisms through which aquatic vegetation regulates the clear–turbid state transition in large, shallow lakes remain poorly understood, particularly the ecological response processes to coupled environmental stressors. This knowledge gap impedes the development of precise eutrophication control strategies based on vegetation restoration. To address this, a process-based eco-dynamic model for submerged aquatic vegetation (the WET model) was developed. Integrating long-term observational data, the model couples

hydrological processes, nutrient biogeochemical cycles, and vegetation dynamics. This framework facilitates the identification of key ecological thresholds and provides a theoretical and methodological foundation for transitioning from qualitative description to quantitative prediction in the ecological restoration of large, shallow lakes. This study addresses these challenges through three primary objectives: (1) to analyze the spatiotemporal patterns of aquatic vegetation cover in Lake Taihu over a 15-year period and identify its key drivers using long-term time-series data; (2) to establish the relationship between hydrological/physicochemical drivers and submerged vegetation response in Lake Taihu using the submerged vegetation eco-dynamics (WET) model; (3) to quantify the nonlinear response and control thresholds of aquatic vegetation cover to eutrophication in large, shallow lakes by evaluating N:P ratio-based indicators across multiple eutrophication scenarios. These objectives are achieved through scenario-based modeling of the Lake Taihu ecosystem, elucidating the critical role of vegetation cover in eutrophication dynamics. The findings provide a scientific foundation for developing targeted vegetation restoration strategies to control eutrophication. Furthermore, they offer critical insights into eutrophication recovery management and elucidate the mechanisms governing the macrophyte-to-algae regime shift in large, shallow eutrophic lakes.

Materials and methods

Study area

Taihu Lake, the third-largest freshwater lake in China, has an average depth of 1.9 m. It is classified as a typical shallow lake, covering an area of 2338 km² and supporting a basin population of approximately 60 million²⁰. Taihu Lake, located at the junction of southern Jiangsu Province and northern Zhejiang Province in China, lies between latitudes 30°55′40″–31°32′58″N and longitudes 119°52′32″–120°36′10″E. Taihu Lake is located in the heart of the Yangtze River Delta, the most industrialized and urbanized region in China²⁶. Since the 1980s, cyanobacterial blooms have posed a significant environmental challenge in Taihu Lake⁴⁰. Human activities are the primary drivers of accelerated water eutrophication, with agricultural non-point source pollution representing the largest proportion, while point-source discharges from industrial and domestic wastewater collectively elevate nitrogen and phosphorus concentrations⁴¹. Increased eutrophication and degradation of the underwater environment in Taihu Lake have led to the loss of vegetation, thereby disrupting the balance of the lake's aquatic ecosystem and contributing to its continued degradation^{37,42}. The highest density of aquatic vegetation is found in the eastern region of Taihu Lake, predominantly composed of submerged macrophytes, while the southern region supports a mixed community of submerged and floating-leaf macrophytes. In contrast, aquatic vegetation has largely disappeared from the open-water zones of the central and northwestern regions¹⁷.

Data sources and processing

The historical meteorological data for this study were obtained from the Resource Service System (cern.ac.cn), with measurements collected at the Taihu Lake Station (Wuxi Headquarters), including daily records of air temperature, dew point temperature, barometric pressure, wind speed, and cloud cover. Additionally, historical flow data were obtained from the Taihu Lake Basin Management Bureau under the Ministry of Water Resources (MWR), which provided monthly rainfall and discharge data. Water quality data covering the period from 2005 to 2019 were also collected. This dataset includes 15 years of monthly data from 32 sampling points in Taihu Lake, facilitating continuous monitoring of 27 ecological indicators. Key indicators include water temperature, dissolved oxygen, total nitrogen, total phosphorus, and chlorophyll-a. The distribution of sampling points is shown in Fig. 1c, and the samples were analyzed in accordance with the “Technical Specification Requirements for Surface Water and Wastewater Monitoring” (People's Republic of China Ministry of Environmental Protection, 2002). For satellite data, observations were made in February, May, August, and November of each year from 2005 to 2019. The remotely sensed data were acquired from Landsat TM/ETM+/OLI surface reflectance datasets. Field data on the dry weight of aquatic plants were collected biannually from 2000 to 2019 to validate the remotely sensed data.

WET model principles

The WET (Water Ecosystems Tool) model consists of six key modules that work together to simulate various aquatic processes. These modules include the abiotic-water, abiotic-sediment, phytoplankton-water, phytoplankton-sediment, macrophytes, and auxiliary modules⁴³. In addition to these core modules, the model incorporates essential physicochemical processes, such as sedimentation, deposition, resuspension, diffusion, adsorption, mineralization, and burial, which are crucial for accurately simulating aquatic interactions. Primary producers, such as macrophytes and phytoplankton, are central to the model's simulations. Oxygen dynamics, such as oxygen produced by photosynthesis and consumed by mineralization in both the water column and sediment, are also considered, providing a comprehensive understanding of the ecological balance in aquatic environments⁴⁴.

The aquatic plant module primarily simulates submerged plants and incorporates key processes such as photosynthesis, nutrient uptake, respiration, excretion, mortality, migration, and seasonal growth and emergence cycles. These processes are essential for accurately modeling the life cycle and ecological interactions of aquatic plants within the ecosystem⁴³. Figure 2 presents a conceptual diagram of the WET model, adapted from Hu et al.⁴⁴. The left side shows the model's input data, while the right side presents the primary output data. The inflow of phosphate, dissolved and particulate organic phosphorus, ammoniacal nitrogen, dissolved and particulate organic nitrogen, and nitrate nitrogen into the lake acts as a boundary condition. The output data include key indicators such as dissolved oxygen, lake water quality, aquatic plant cover, and water temperature³⁹.

$$\text{Macrophytes dry weight } [\text{gD}/\text{m}^2]$$

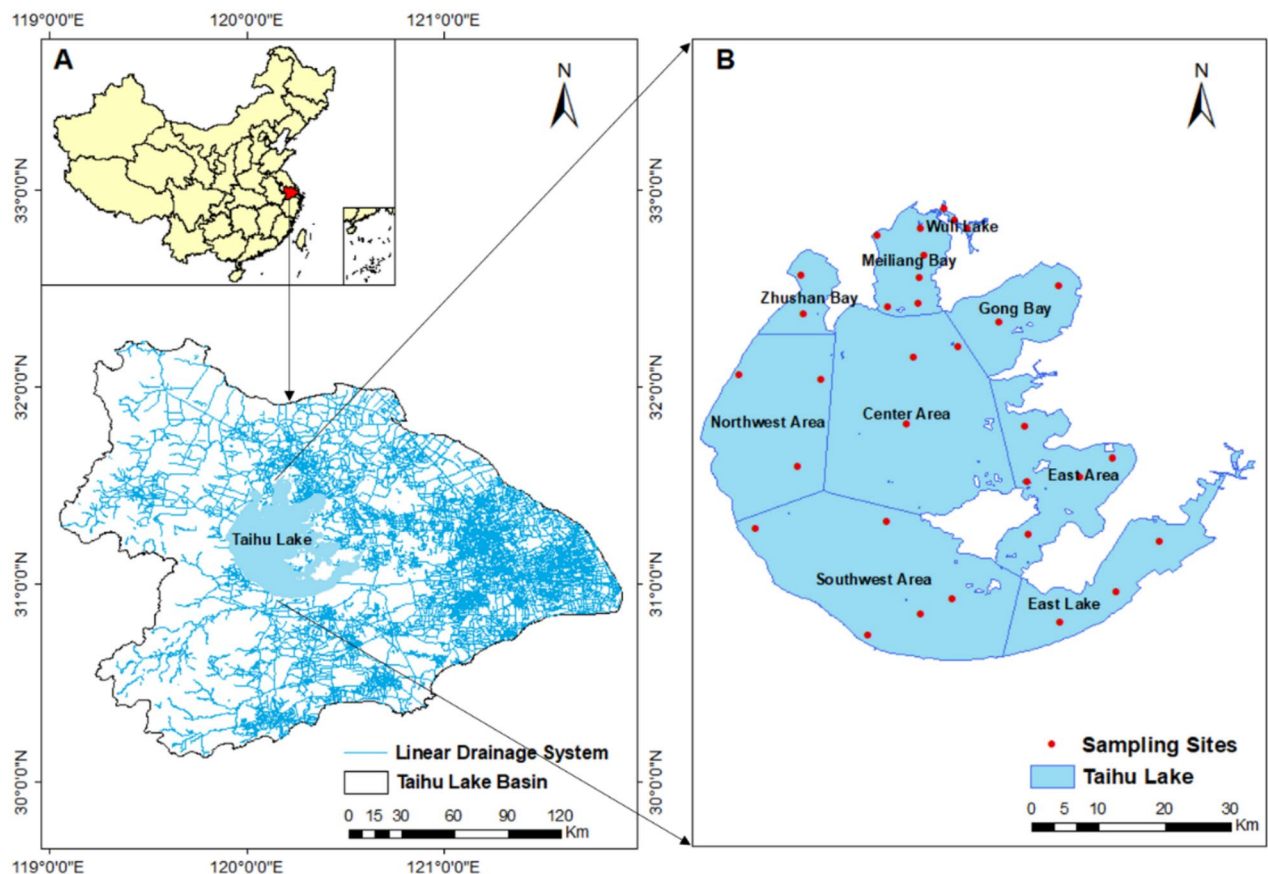


Fig. 1. Study area and sampling sites. (A) geographic location of Taihu Lake in the Taihu Lake basin; (B) Taihu Lake water quality sampling sites and Taihu Lake division.

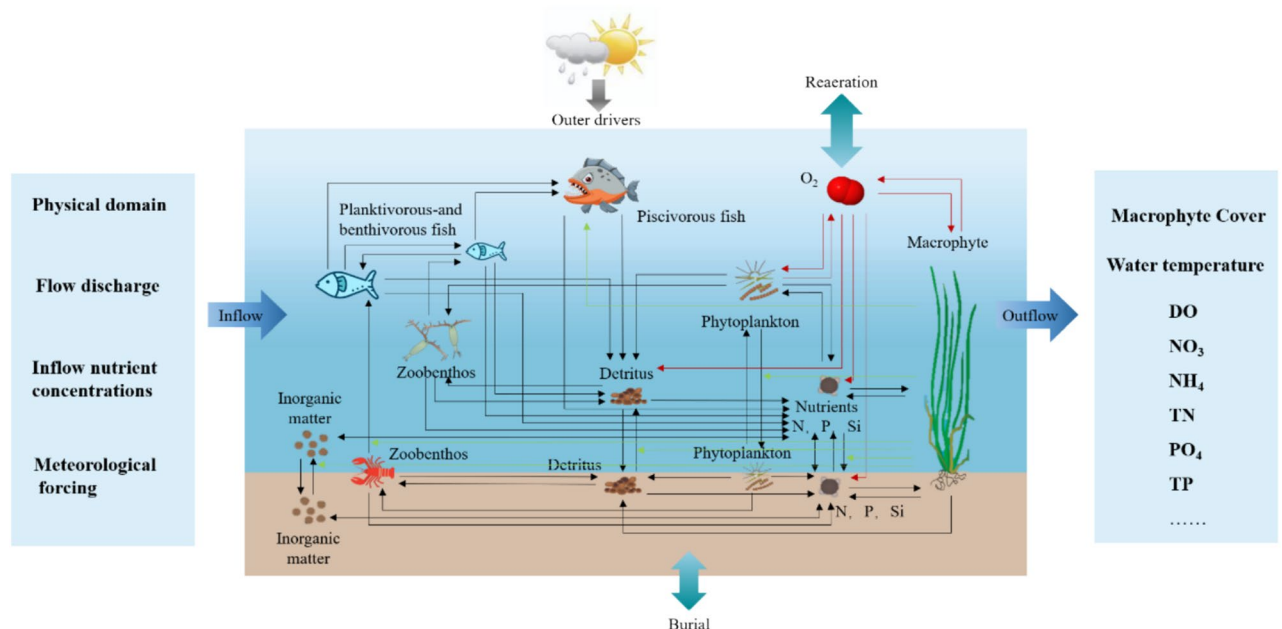


Fig. 2. Conceptual diagram of the aquatic ecosystem model GOTM-FABM-PCLake, adapted from Hu et al.⁴⁴ under a CC BY 4.0 license. Black arrows represent interactions, red arrows depict dissolved oxygen kinetic processes, and green arrows indicate aquatic plant-mediated processes.

$$\begin{aligned} dDVeg &= < tDProdVeg - tDRespVeg - tDMortVeg + tDMigrVeg > \\ &= < Production - Respiration - Mortality + Migration > \end{aligned} \quad (1)$$

The dry weight of the aquatic plant module in the model is calculated using Eq. (1). The net change variable in submerged plant biomass per unit time (dDVeg) equals total primary production (tDProdVeg) minus total respiratory consumption (tDRespVeg) minus total mortality (tDMortVeg) plus net migration (tDMigrVeg). The migration term in Eq. (1) plays a critical role in mediating the spatial redistribution, population maintenance, and disturbance response dynamics of aquatic plant communities. The dry weight of the aquatic plant is equal to the dry weight produced through aquatic plant production, minus the dry weight lost due to respiration and mortality, plus the dry weight generated during the migration process.

$$R^2 = 1 - \frac{\sum_i (\hat{y}_i - y_i)^2}{\sum_i (\bar{y}_i - y_i)^2} \quad (2)$$

$$RMSE = \sqrt{\frac{1}{m} \sum_{i=1}^m (y_i - \hat{y}_i)^2} \quad (3)$$

The formulas for calculating the metrics used for calibration and validation in the model are given in Eqs. (2) and (3). y_i denotes the actual observed value for the i -th sample, $\hat{y}_i$ denotes the model-predicted value for the i -th sample, $\bar{y}_i$ denotes the mean of all observed values y_i , and m is the sample size. The coefficients of determination (R^2) and root mean square errors (RMSE) are calculated after normalizing the data.

The Pettitt test is a nonparametric method used to detect change points in time series data, utilizing rank-based sequential analysis to identify abrupt shifts in the mean or distributional properties of a dataset.

$$U_K = \sum_{i=1}^k R_i - \frac{k(n+1)}{2} \quad (4)$$

In the Pettitt test, R_i denotes the rank of the data, and k represents the candidate change point position. The test statistic $K = \max |U_K|$, and its statistical significance is evaluated against the asymptotic distribution:

$$p \approx 2 \exp \left(-\frac{6K^2}{n^3 + n^2} \right) \quad (5)$$

If $p < 0.05$, the null hypothesis of homogeneity is rejected, and a statistically significant change point is identified at $\arg \max |U_k|$.

Eutrophication scenario construction in Taihu Lake

To assess the impact of external nutrient loading on aquatic ecosystems, this study used the years 2015–2019 as a baseline for analyzing year-to-year changes in the modeled scenarios. Scenario changes were made by adjusting the boundary and input conditions for the baseline years. The N:P ratios can, to some extent, characterize the eutrophic status of lakes. As the chlorophyll-*a* concentration increases, the N:P ratio significantly decreases. The corresponding mean N:P ratios are 54.1, 26.8, and 18.1 when the lake is in oligotrophic, eutrophic, and hyper-eutrophic states, respectively⁴⁵. These values were used as boundary conditions to simulate different eutrophication states.

To investigate the ecosystem response to nutrient changes, two distinct regulatory scenarios were developed: one for phosphorus regulation and one for nitrogen regulation. Scenario boundaries were defined using the observed minimum and maximum concentrations of TN, TP and the resulting N:P ratios from historical monitoring data at Lake Taihu (2005–2019). For the P-regulation scenario, TP concentration was varied from 0.052 to 0.332 mg/L in 0.005 mg/L increments, while TN was held constant at its historical average (2.859 mg/L). This generated 56 scenarios with N:P ratios ranging from 9 to 57. For the N-regulation scenario, TN was varied from 1.527 to 7.927 mg/L in 0.1 mg/L increments, with TP fixed at its historical average (0.129 mg/L), yielding 65 scenarios with N:P ratios between 12 and 61. All scenarios were simulated over a 60-year period (2020–2089) to ensure the lake ecosystem reached a stable, dynamic equilibrium. Model outputs for aquatic plant cover and nutrient concentrations were evaluated based on summer averages, as this season best represents peak biological activity and the critical conditions for regime shifts. The N:P ratios served as a key integrative indicator to simulate eutrophication status under varying levels of aquatic plant cover, thereby providing a quantitative basis for management recommendations (Table 1).

Results

Characterization of spatial and temporal changes in nutrients and aquatic vegetation cover in Taihu Lake

Analyzing long-term trends in N:P ratios and aquatic plant cover is crucial for understanding the ecohydrological dynamics of Lake Taihu. The average interannual N:P ratio in Lake Taihu exhibited an overall decreasing trend from 2000 to 2019. This period began with an increase from 19.5 in 2000 to a peak in 2003, followed by a decline until 2010. A secondary peak then occurred before ratios declined again through 2019. This overall decline in the N:P ratio suggests a general deterioration in trophic status, consistent with the association of lower N:P ratios with more severe eutrophication in this system. Similarly, aquatic plant cover, which is predominantly

Control (P/N)	Index	Min	Max
P Control (Increment: 0.005)	TP (mg/L)	0.052	0.332
	TN (mg/L)	2.859	
	N:P ratios	57	9
N Control (Increment: 0.1)	TP (mg/L)	0.129	
	TN (mg/L)	1.527	7.927
	N:P ratios	12	61

Table 1. Scenario settings for two conditions of nitrogen and phosphorus regulation to control N:P ratios.

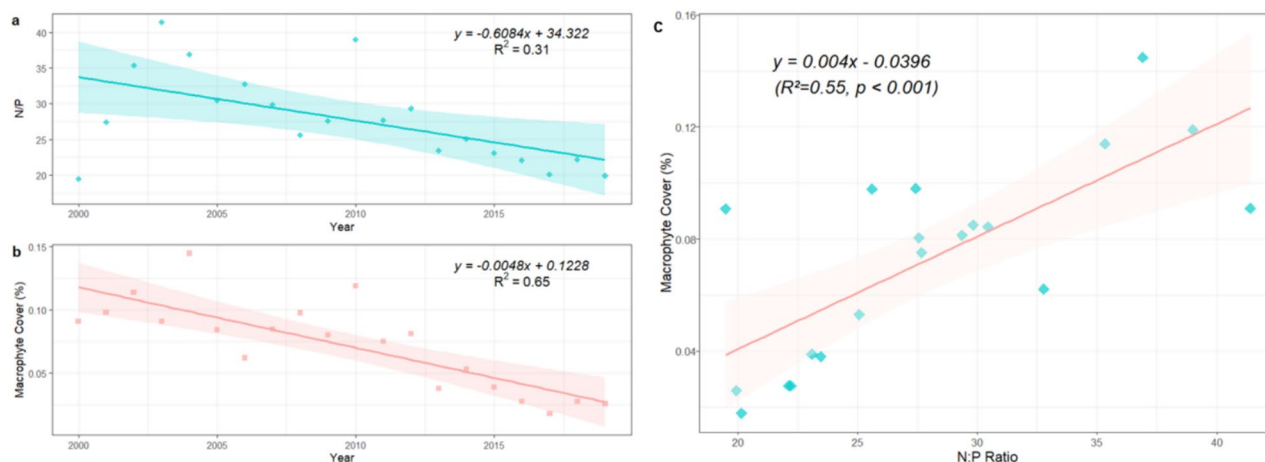


Fig. 3. Trends in aquatic plants and N:P ratios. (a) Average annual trends in N:P ratios from 2000 to 2019; (b) Average annual trend of aquatic plant cover from 2000 to 2019; (c) Correlation analysis of interannual nitrogen-phosphorus ratio and aquatic plant coverage from 2000 to 2019.

submerged vegetation, displayed a decreasing trend. A particularly sharp decline commenced in 2015. This abrupt decline coincided with the initiation of centralized, high-intensity mechanical salvage operations. These operations directly removed aboveground biomass and fragmented rhizome networks, thereby impairing regenerative capacity. Concurrently, the repeated disturbance reduced sediment seed banks and increased water turbidity, which further inhibited seedling establishment and vegetative regrowth¹⁷. A highly significant positive correlation was identified between N:P ratios and aquatic plant cover in Lake Taihu ($p < 0.001$), reflecting their synchronous temporal dynamics. These correlated long-term trends are essential for understanding the ecohydrological dynamics of Lake Taihu. The analysis confirmed a significant decreasing trend in the average interannual N:P ratio over the study period (Fig. 3).

The spatial patterns of aquatic plants and N:P ratios in Taihu Lake were analyzed using data from 30 monitoring points for aquatic plant dry weight and 32 hydrological monitoring stations. The distribution patterns of both variables across the divisions of Taihu Lake were examined. The spatial distribution of aquatic plant dry weight is shown in Fig. 4a, with the highest dry weight found in the East Lake and East Area. The concentration decreases from the southeast to the northwest. The N:P ratios of Taihu Lake, shown in Fig. 4b, exhibit a similar spatial trend, increasing from the southwest to the east. The N:P ratios are higher in the eastern part of the lake and lower in the western part. The spatial trends of both variables generally follow the same pattern from the Southwest Area to the Eastern Area, showing a decreasing trend as illustrated in Fig. 4c. The average values of sampling points in each region were calculated to analyze the N:P ratios and dry weight of aquatic plants in the Southwestern Area, Gong Bay, Eastern Area, and East Lake. The concentrations of both indicators were highest in the Eastern Area. Both indicators exhibited a linear increasing trend from the Southwestern Area to the Eastern Area, demonstrating similar spatial changes.

Analysis of spatiotemporal patterns revealed a consistent coupling between N:P ratios and aquatic plant biomass in Lake Taihu. This coupling indicates that aquatic vegetation is a significant driver of N:P ratios, which in turn governs the eutrophication status of the lake.

Construction and validation of the WET eco-dynamic model for large shallow lakes

An ecological dynamics model for a large, shallow lake was developed to simulate the complex input–output relationships between nutrients and aquatic vegetation in Taihu Lake. The dataset spans from 2005 to 2019, with data from January 2005 used as the initial condition for the model. The model runs for three years as a warm-up phase to minimize calibration uncertainty, with the period from January 2008 to December 2014 designated as the calibration period and from January 2015 to December 2019 as the validation period. Parameters were

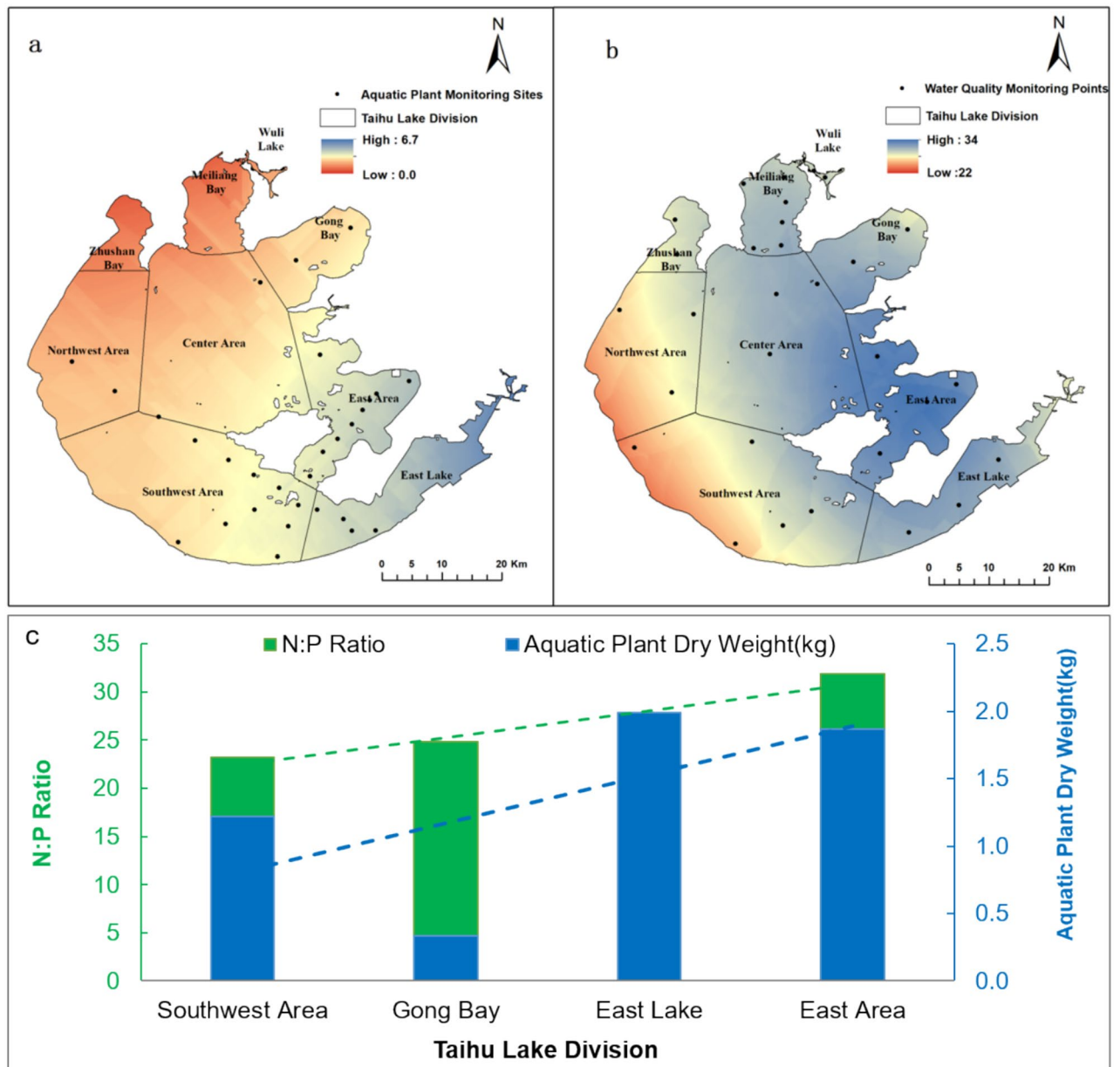


Fig. 4. Response pattern of spatial distribution of dry weight and N:P ratios of aquatic plants. (a) Spatial interpolation results of dry weight concentration at 30 aquatic plant dry weight sampling points in Taihu Lake; (b) Spatial interpolation results of nitrogen and phosphorus ratios at 32 water quality sampling points in Taihu Lake; (c) Histogram of the average of sampling points in four sub-districts: Southwest Area, Gong Bay, East Lake, and East Area.

adjusted to achieve optimal model performance for water quality variables, based on previous sensitivity analyses and the modeling expertise of Janse et al.⁴⁶. The model was calibrated and optimized using the ACPy auto-calibration tool, which employs a differential evolution algorithm to maximize the likelihood function. Model performance was evaluated by calculating the Root Mean Square Error (RMSE) and the coefficient of determination (R^2) through a comparison of simulated outputs against observed data. The calibration process iteratively adjusted parameters until these metrics reached their optimal values and the model output stabilized. The final calibrated parameter values are reported in Supplementary Material, Table S1.

The model was calibrated and validated using water quality, meteorological, and hydrological data from Taihu Lake, spanning the period from 2008 to 2019. For water quality, field-measured monthly data of TP, TN, and ammonia nitrogen from monitoring sites in Taihu Lake, covering the same period, were utilized. In Fig. 5, black solid points represent the validation values during calibration, while black hollow points indicate the validation values during the validation phase. The simulation is depicted by a black line, with a 15% error band illustrating the uncertainty associated with the observations. The model-simulated TN closely matched the observed values, as shown in Fig. 5a, with a deviation of less than 20% and an R^2 greater than 0.6. However, the

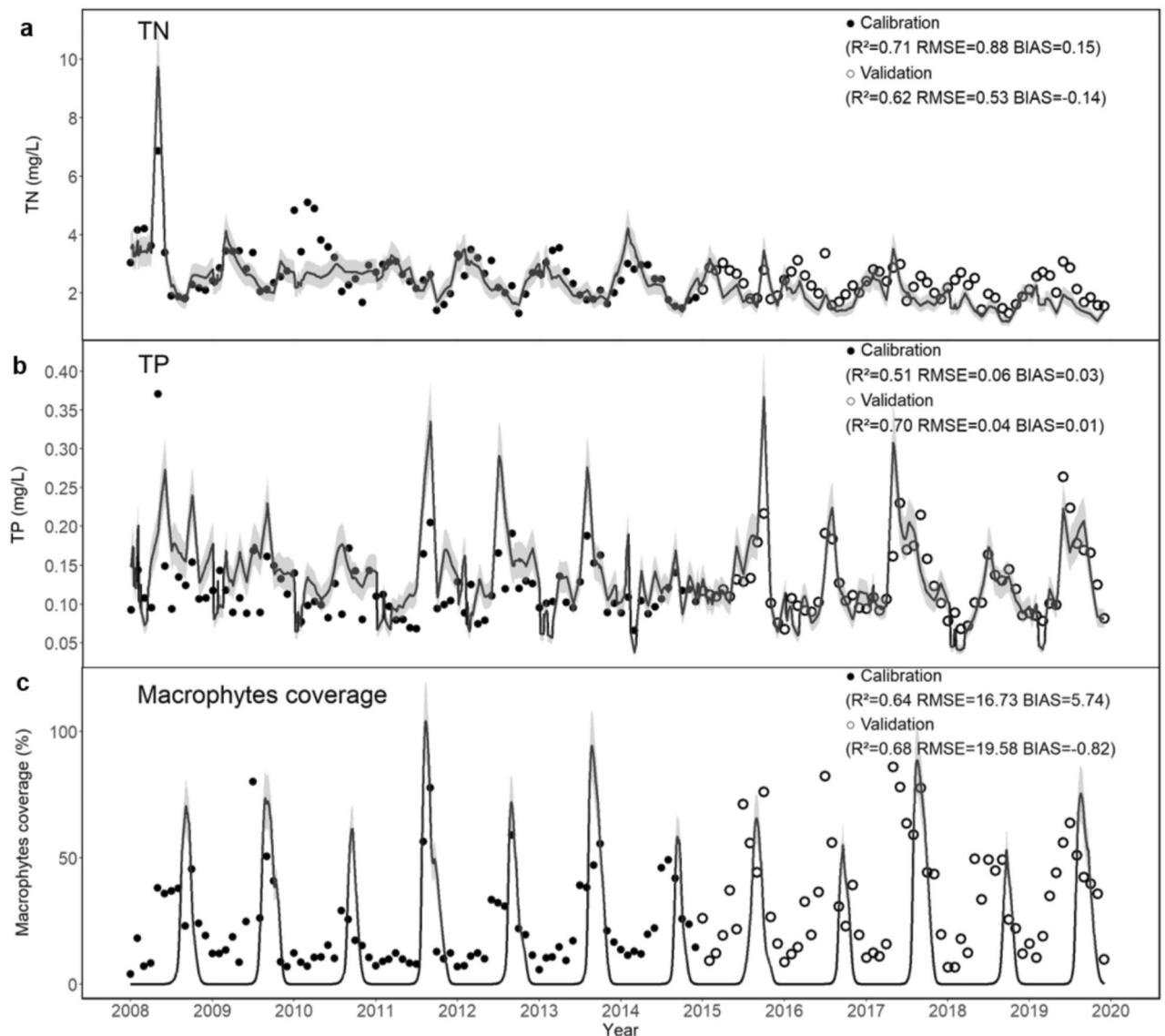


Fig. 5. Model simulated values (black line) and calibration and validation observations (black solid and white hollow dots, respectively). Gray bands represent the uncertainty associated with the 15% error introduced into the indications and observations (a) TN; (b) TP; (c) Macrophytes coverage.

model did not consistently capture the maximum and minimum TN values, indicating limitations in simulating the seasonal peaks of the nitrogen indicator. However, the overall trend was adequately represented. For aquatic organisms, the model successfully captured the overall trend in aquatic plant cover, showing a decrease as winter temperatures cooled. However, the model overestimated aquatic plant cover during the spring and summer of 2011. The model generally describes submerged plant cover levels; however, the simulated cover was typically lower than the observed values during the fall and winter months. This discrepancy may be attributed to limited light availability during these seasons, which results in reduced simulated growth. This suggests that the water quality and aquatic plant simulations are accurate and reliable, demonstrating the overall applicability of the WET model to Taihu Lake. The simulation results meet the requirements of this study.

Nonlinear response process of aquatic plant cover in Taihu Lake under different eutrophication states

Using the constructed ecological kinetic mechanism model of Taihu Lake, a total of 121 scenarios for nitrogen and phosphorus regulation were simulated. Process curves and inflection points for aquatic vegetation in Taihu Lake under varying nitrogen and phosphorus inputs were derived, as shown in Fig. 6. When the Chlorophyll-*a* levels in the lake indicated oligotrophic, eutrophic, and hypereutrophic states, the corresponding mean N:P ratios were 54.1, 26.8, and 18.1, respectively⁴⁵, thus classifying the lake into three states based on the grading of N:P ratios. The pink curve in Fig. 6 represents the 56 scenarios where N:P ratios vary from 12 to 61 by adjusting the TN concentration between its minimum and maximum values (1.527 to 7.927 mg/L), while maintaining

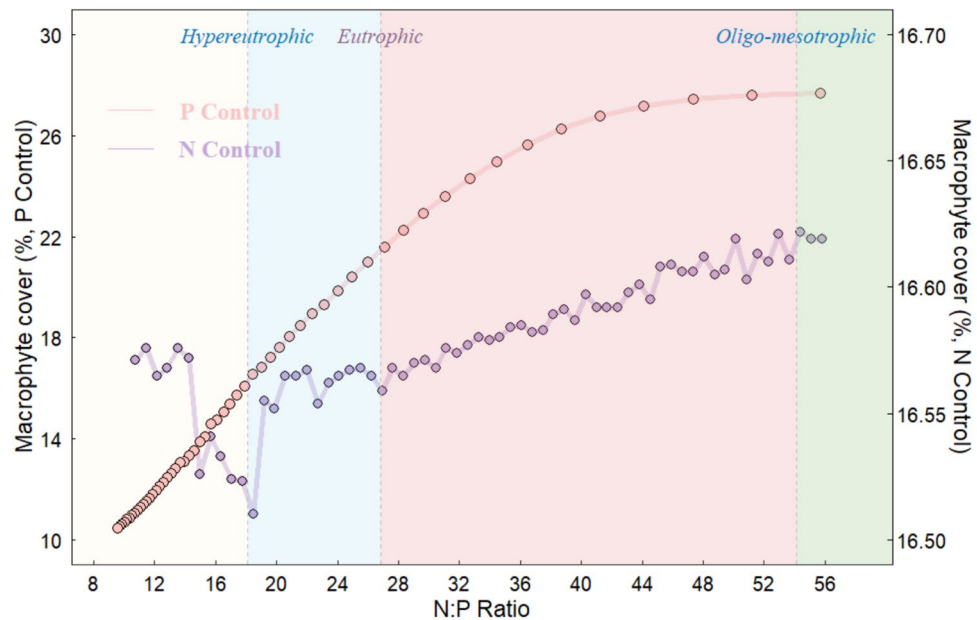


Fig. 6. Scenario simulation of aquatic plants regulating the dynamic response process of lake eutrophication.

a constant TP concentration at the monthly average of 0.129 mg/L from 2005 to 2019. The flat purple curve represents the 65 scenarios in which the TN concentration was kept constant at the average monthly value of 2.859 mg/L over the past 15 years, while the TP concentration varied between its minimum and maximum values (0.052 to 0.332 mg/L), resulting in N:P ratios ranging from 9 to 57. The load map in Fig. 6 depicts the average aquatic plant cover during the summer months of the last five years of the scenario simulation. In the nitrogen control scenario, changes in aquatic plants show a slight increase in the N:P ratio to 14.3 after the aquatic plant cover declines to 16.51%. Beyond this point, as aquatic plant cover continues to rise, the change interval remains approximately 0.1%. The growth of aquatic plant cover in Taihu Lake is less restricted by nitrogen than by phosphorus. The impact of nitrogen is minimal, resulting in an overall non-significant trend that levels off. In contrast, changes in aquatic vegetation under phosphorus control show a nonlinear increase and are more sensitive than those under nitrogen control.

The results of the Pettitt mutation test for the phosphorus control simulation scenario indicated that the inflection point for changes in aquatic plant cover occurred at 15.28%. Under phosphorus control, increasing aquatic plant cover in the lake effectively mitigates the eutrophic state. As aquatic plant cover rises to 27.7%, the lake's N:P ratio reaches a point where it transitions to a mesotrophic state, signaling an improvement in eutrophication. Aquatic plants play a crucial role in aquatic ecology. Expanding the area of submerged plants in the lake can help alleviate eutrophication.

Discussion

Temporal and spatial patterns of changes in nutrients and aquatic vegetation in large shallow lakes and their impact relationships

The spatiotemporal dynamics of aquatic vegetation cover and N:P ratios in Taihu Lake exhibit remarkable consistency. A severe cyanobacterial bloom occurred in Taihu Lake in 2007, followed by repeated bloom-induced eutrophication, with N:P ratios gradually decreasing⁴⁷. Data analysis indicates that the aquatic vegetation cover in Taihu Lake exhibited an overall decreasing trend with interannual variations. By 2015, the cover had been reduced to about 5%, consistent with trends observed in remote sensing data from Taihu Lake between 1980 and 2017. As shown in Fig. 4b, the N:P ratios in Taihu Lake exhibited a decreasing trend from east to west⁴⁸. Additionally, the water quality in the western part of the lake was more polluted than in the eastern part, consistent with findings from previous studies. The injection of large nutrient loads into the northwestern part of Taihu Lake due to human activities, combined with the closed geomorphological features in the northern region, is the primary factor contributing to the spatial heterogeneity in water quality⁴⁹. As shown in Fig. 4a, the aquatic vegetation cover gradually decreased from the eastern part to the middle and western sections of Taihu Lake⁵⁰. Remote sensing results revealed that aquatic vegetation was primarily concentrated in the eastern and southeastern regions¹⁷, consistent with the trend observed in this study.

A significant relationship was observed between aquatic vegetation and the trophic status of Taihu Lake. Persistent water eutrophication and changes in hydrological conditions contribute to the ongoing decline of aquatic vegetation in Taihu Lake⁵⁰. The large-scale reduction of aquatic vegetation has disrupted the ecological balance and steady state of the lake, further contributing to the degradation of the water ecosystem²⁶. Spatially, the western part of the lake is dominated by plankton, with sparse aquatic vegetation and frequent algal blooms^{12,13}, whereas the eastern part of the lake is characterized by dense aquatic vegetation and relatively good water quality⁵¹. Conversely, the interaction between water eutrophication and aquatic vegetation demonstrates that

nutrients in hyper-eutrophic lakes lead to the deterioration of aquatic vegetation and the subsequent degradation of lake ecosystems¹⁵. There is a discernible relationship between aquatic vegetation and eutrophication in Taihu Lake, which supports our hypothesis. However, simulating this response requires a more complex analytical model, and the WET model may be more effective for scenario prediction in addressing this issue.

Model construction and advantages of aquatic vegetation in the shallow lake Taihu Lake

Aquatic ecological models are primarily used to study eutrophication, bioaccumulation, and trophic interactions in aquatic ecosystems⁵². Several large-scale aquatic ecosystem models, including PCLake, WET, EFDC, CEQUAL-W2, WASP, DELFT3D, AQUATOX, and MIKE, are well-developed and commercially available⁵³. The WET model is an aquatic ecosystem model that allows flexible, modular configuration of the food chain, making it suitable for a wide range of aquatic ecosystem applications. It integrates hydrodynamic, biogeochemical, and ecological processes to comprehensively simulate lake ecosystems, with multiple modules capable of simulating complex ecosystem interactions. This complexity enables a more accurate simulation of changes in aquatic vegetation⁴³. The WET model can simulate biotic and abiotic transport and transformation processes, as well as climate and nutrient loading change scenarios in a lake ecosystem. For example, Nielsen et al. applied the model to Arreskov Lake, a shallow Danish lake, to predict the impacts of future climate change and nutrient loading on lake water quality. Chen et al. used the WET model to assess the impacts of extreme climatic events on eutrophication levels in Bryrup Lake, Denmark⁵⁴. Similarly, Andersen et al. employed the WET model to predict changes in ecosystem states of shallow lakes³⁸.

The WET model provides mechanistic explanatory capabilities for simulating aquatic vegetation in lakes and aligns with the scientific principles governing growth and changes in aquatic biodynamic processes. It incorporates the vertical gradient from the hydrodynamic model GOTM, analyzed in conjunction with the aquatic ecosystem model PCLake, and is implemented within the framework of Aquatic Biogeochemical Modeling (FABM). This approach more accurately models the delayed growth and cumulative effects of aquatic vegetation, better aligning with the observed phenomenon in which aquatic vegetation does not decline immediately after an algal bloom outbreak. Compared to using PCLake alone, the WET model offers better lag identification capabilities and more accurately reflects the actual conditions of the lake. During practical validation, the WET model successfully simulates the coverage and growth processes of aquatic vegetation in shallow lakes. Phytoplankton limitation by nutrients is modeled using the Drope equation, which also accounts for oxygen dynamics, including reaeration, oxygen production via photosynthesis, oxygen exchange with the water column, and oxygen consumption during sediment mineralization³⁸. The effect of temperature on aquatic vegetation growth can be represented by two exponential functions, which describe its impact on production respiration and maintenance respiration, respectively³⁴.

Nonlinear inflection thresholds and cause identification for aquatic vegetation restoration and eutrophication scenarios in shallow lakes

The increase in aquatic vegetation coverage to control lake eutrophication and inhibit algal blooms aligns with the results of the scenario simulation presented in this study⁵⁵. Phosphorus limitation in lake ecosystems is governed by its distinct biogeochemical behavior at the sediment–water interface and its long-term accumulation. Aquatic vegetation exhibits a pronounced uptake capacity for phosphorus, which is sustained by a dynamic release–uptake cycle at the sediment interface that maintains a stable recycling pathway. In contrast, nitrogen is typically non-limiting due to significant gaseous losses via denitrification, constraints on biological fixation, and complex transformations among its various chemical species. These processes collectively result in lower and less predictable nitrogen bioavailability compared to phosphorus. The Pettitt mutation test indicated that the curve's inflection point occurred when aquatic vegetation cover reached 15.28%, at which point the lake entered a state of hyper-eutrophication in response to the simulated phosphorus. The results of the Pettitt mutation test are presented in Fig. S3 of the Supplementary Material, indicating that increasing aquatic vegetation cover to 15.28% is insufficient to improve the health of Taihu Lake. However, increasing the aquatic vegetation cover to 27.7% resulted in an increase in the N:P ratio to 56.1, causing the lake to transition from a hyper-eutrophic state to an eutrophic one and ultimately to an oligotrophic state. The 27.7% threshold aligns with the lower bound of the optimal range identified in the Baiyangdian study, likely indicating that a self-sustaining, resilient clear-water state has been established⁵⁶.

The model simulates the varying effects of aquatic vegetation on the dynamics of nitrogen and phosphorus. Phosphorus absorption by aquatic vegetation increases linearly, while nitrogen absorption is more sensitive to vegetation, initially declining before increasing. In lakes, phosphorus uptake begins with the direct absorption of phosphate, followed by the accumulation of phosphorus as aquatic plants develop. As aquatic vegetation wilts and decays, phosphorus stored in plant tissue is released back into the water, causing a short-term increase in phosphorus concentration. However, this process is generally slow, with phosphorus changes occurring as a long-term, steady-state increase. As aquatic vegetation decays, phosphorus is released back into the water. Additionally, the root system of aquatic vegetation plays a role in influencing phosphorus dynamics in the sediments⁵⁷. Additionally, increased aquatic vegetation cover enhances water transparency and reduces plankton growth, thereby indirectly limiting phosphorus cycling. The influence of aquatic vegetation on nitrogen is primarily associated with its polymorphic nature and complexity. The scenario simulation of the WET model currently simplifies ecological processes, with the simulation results mainly reflecting the response characteristics driven by nitrogen-phosphorus structure changes. Due to data acquisition limitations and the preliminary positioning of the model, the current framework is simplified. Future research will incorporate remote sensing technology and water quality monitoring data to supplement variables such as transparency, water depth, and other key factors, ultimately building a more ecologically complete multi-factor coupling model to enhance both its ecological adaptability and universality.

Moreover, submerged vegetation plays a crucial role in maintaining clear water conditions in lakes through various mechanisms⁵⁸. These plants reduce turbidity and nutrient concentrations, including total nitrogen and phosphorus, by acting as a barrier to particle dispersal and filtering suspended particles from the water. Submerged plants also directly absorb nitrogen and phosphorus through their roots and leaves, thereby improving water quality by lowering the concentrations of soluble nitrogen and phosphorus⁵⁰. Additionally, by secreting allelopathic substances, they modify algal cell physiology, inhibiting phytoplankton growth and reducing algal biomass. The leaf type of submerged plants plays a crucial role in enhancing water quality. Leaf shape is a key phenotypic trait that reflects their adaptive strategies to environmental stress. Establishing a submerged plant system not only improves water transparency, but the resulting clarity further supports plant growth, creating a positive feedback loop. It provides food for microorganisms and shelter for zooplankton, reducing suspended nutrients and inhibiting algal growth through zooplankton grazing⁵⁹. Therefore, large-scale cultivation of submerged vegetation in lakes can improve aquatic ecosystem health and help mitigate eutrophication in Taihu Lake. In the future, emphasizing the ecological role of aquatic vegetation may alleviate the burden of lake management.

This study establishes a 27.7% aquatic vegetation coverage threshold for transitioning Lake Taihu from a eutrophic to an oligotrophic-mesotrophic state, providing a quantifiable management target that bridges scientific research and practical application. This threshold serves as a key indicator for assessing ecosystem status and restoration effectiveness. The WET model serves as a predictive decision-support tool, simulating vegetation response to management actions—including pollutant load reduction and hydrological regulation—to optimize strategies and forecast eutrophication trends. Furthermore, these findings enable the development of an early-warning system that triggers intervention when vegetation coverage nears critical levels, facilitating a shift from reactive treatment to proactive, preventative management.

Conclusions

This study developed a novel WET-based ecological model for large, shallow lakes, establishing a nonlinear relationship between long-term hydrological/physicochemical drivers and submerged vegetation cover in Lake Taihu. A key contribution is the quantification of critical ecological thresholds, demonstrating that achieving approximately 27.7% vegetation cover can shift the system from a eutrophic to an oligotrophic-mesotrophic state under specific N:P ratios. The results confirm phosphorus as the primary limiting nutrient, with nitrogen exerting a secondary influence on vegetation dynamics. A primary limitation is the model's parameterization for a single lake system, necessitating validation across diverse lacustrine environments to ensure broader applicability. Future work should elucidate the underlying biogeochemical mechanisms—particularly sediment-phosphorus feedbacks and nitrogen fixation—and evaluate the impacts of climate change on restoration resilience. For management, this study provides quantifiable vegetation targets and a predictive tool to simulate ecosystem responses to interventions, thereby enabling proactive eutrophication control and optimized resource allocation in shallow lakes.

Data availability

All data supporting the findings of this study are included in the manuscript. The datasets used and analysed during the current study available from the corresponding author on reasonable request.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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